Supplementary Material for

An Efficient Mixed-Integer Linear Programming Formulation for Solving Influence Diagrams

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1. Introduction

This file contains additional computational examples for the manuscript "An Efficient Mixed-Integer Linear Programming Formulation for Solving Influence Diagrams". The first example is a variation of the extended oil wildcatter problem in which earlier test results are known when commissioning further tests. The second computational example is a variation of the pig farm problem (Lauritzen and Nilsson, 2001) whose structure suits the Rooted Junction Tree (RJT) formulation. Finally, the last Section presents a variant of the turbine inspection and maintenance problem, which is solved by discretizing continuous variables.

The RJT-models and the DP formulation were constructed using the Julia package DecisionProgramming.jl v2.0.1. The experiments were conducted using an Intel E5-2680 CPU at 2.5 GHz with 16 GB of RAM, provided by the Aalto University School of Science's "Science-IT" project. The code and data to support the numerical experiments in this supplementary material can be found via https://github.com/toubinaattori/Decision-Programming-new-formulation. To solve the optimization models, the Gurobi solver was used with default parameters and parameters specified in Table 1.

Table 1: Solver parameter settings in the experiments.

Parameter	Value	Description
TimeLimit	3600	Limit total runtime (1h)
Threads	1	Thread count
Presolve	0	Disable presolve
Heuristics	0	Minimize time spent in MIP heuristics
NodeMethod	1	Use dual simplex for MIP node relaxations
Cuts	1	Moderate cut generation
MIPFocus	2	Focus on proving optimality

2. Additional computational experiments

2.1. Sequential extended oil wildcatter problem

Consider a variant of the extended oil wildcatter problem, where there is a possibility to carry out at most M tests sequentially, so that previous test results (or the lack of results) inform

Table 2: Nodes of the extended oil wildcatter problem.

Node	Type	States	Explanation
O	Chance node	dry, medium, wet	The state of the oil well
S	Chance node	dry, medium, wet	Geological features of the drilling site
T_i	Decision node	yes, no	Decision to commission test i
R_i	Chance node	N/A, dry, medium, wet	Results of test i
C_i	Value node	-	Cost of test i
D	Decision node	yes, no	Drilling decision
U	Value node	-	Reward minus drilling costs

further testing decisions. The influence diagram of the sequential extended oil wildcatter problem with two tests is represented in Figure 1. The nodes of the problem are presented in Table 2. Because this influence diagram is obtained from the non-sequential one by adding arcs from earlier test results to later testing decisions, every non-sequential strategy corresponds to a sequential strategy that ignores the earlier results. Hence, this variant introduces more flexibility in determining the testing portfolio, which may lead to lower testing costs. However, in the sequential variant the testing period may take more time, which can increase other related costs such as those related to workforce and testing equipment. Therefore, decision makers should consider these tradeoffs when determining which variant of the problem to model.

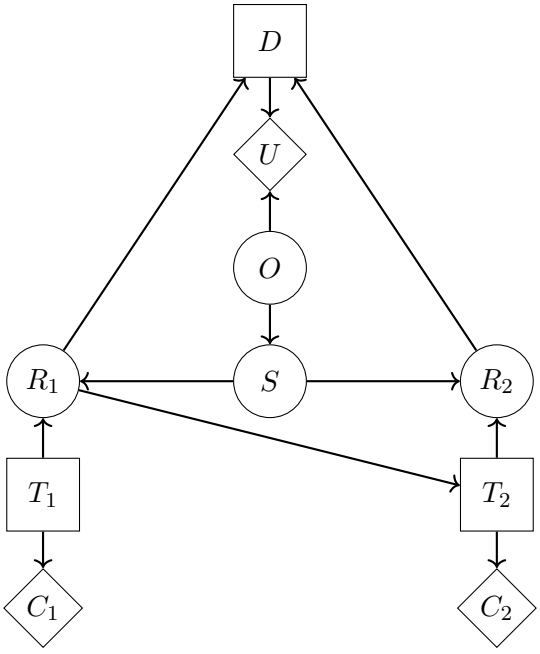


Figure 1: Influence diagram of the sequential extended oil wildcatter problem, where $M = 2$.

As in the previous experiments, 50 random instances of the sequential extended oil wildcatter problem are generated for each $M \in \{1, \ldots, 8\}$. Probabilities and utilities are generated as in the extended oil wildcatter experiment, using the weight vector (equation 13 in the main manuscript) with the same parameter values, and setting the probability of N/A at the test result node to 1 if the test is not carried out and 0 otherwise. The generated problems are then solved with the RDP, RJT, and DP formulations.

Table 3: Results of the sequential extended oil wildcatter problem with 50 analyzed instances per problem type using the default Gurobi parameters.

	RDP						RJT						DP					
M	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %	# c	# v	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %	# c	# v	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %	# c	# v
1	0.01	0.00	50/50	+0	19	26	0.01	0.00	50/50	+3.3	196	144	0.07	0.00	50/50	+121	15	154
2	0.39	0.01	50/50	+67	79	170	2.47	0.01	50/50	+0	795	610	1.38	0.12	50/50	+365	63	1194
3	0.53	0.03	50/50	+165	319	1194	2.83	0.09	50/50	+0	3170	2468	3.87	2.27	50/50	+918	255	9386
4	1.23	0.71	50/50	+366	1279	8874	5.44	0.65	50/50	+0	1e4	9894	129.9	126.5	50/50	+2161	1023	7e4
5	285.0	283.1	16/21	+1368	5119	7e4	50.60	7.64	50/50	+0	5e4	4e4	400.5	371.8	0/0	N/A	4095	6e5
6	T	T	0/0	N/A	2e4	5e5	783.0	45.92	49/49	+0	2e5	2e5	T	T	0/0	N/A	2e4	5e6
7	T	T	0/0	N/A	8e4	4e6	T	T	0/0	N/A	8e5	6e5	T	T	0/0	N/A	7e4	4e7
8	T	T	0/0	N/A	3e5	3e7	T	T	0/0	N/A	3e6	3e6	T	T	0/0	N/A	3e5	3e8

Tables 3 and 4 present the results of the computational experiments. Even though the number of variables and constraints increases only marginally, the computation times increase dramatically compared to the non-sequential oil wildcatter problem. In this experiment, the RJT formulation is the most efficient for larger influence diagrams, achieving the fastest total times in instances with $M \geq 5$ tests under the default Gurobi parameters, and with $M \geq 4$ tests under the modified parameters. The RJT formulation finds the optimal decision strategy in nearly all instances with $M = 6$ tests within the time limit, unlike the other formulations that find a feasible solution within the time limit but fail to converge to the optimal solution. Instances of size $M \geq 7$ cannot be solved with any of the tested formulations.

These computational results show that the RDP and DP formulations are especially sensitive to the number of decision strategy variables. In the non-sequential extended oil wildcatter problem, the number of drilling decision strategies grows exponentially, but the number of testing strategies grows only linearly with respect to the number of tests. In the sequential extended oil wildcatter problem, the number of possible drilling decision strategies is equivalent to that in the non-sequential case, but the number of possible testing strategies grows exponentially with respect to the number of available tests. For instance, in the non-sequential case with $M = 6$ tests, the number of drilling strategy variables is approximately 8000, whereas the number of testing strategy variables is 12. In the sequential case, the number of possible testing strategies jumps to 2730, which is two orders of magnitude more than in the non-sequential case. The RJT formulation, on the other hand, is largely unaffected by the number of decision strategy variables. This example highlights that the RDP formulation is best suited for influence diagrams with relatively few decision nodes and relatively many chance nodes.

2.2. Pig farm problem

The pig farm problem (Lauritzen and Nilsson, 2001) is a classical influence diagram problem whose structure suits the RJT formulation (Herrala, Terho and Oliveira, 2025). In this problem, a pig farmer raises pigs for a four-month breeding period, and subsequently sells them for profit. Each month during the breeding period, the pigs may contract a disease, which ultimately reduces their retail price. During the breeding period, a veterinarian makes monthly visits to the farm and tests the pigs for the disease. Based on the test results, the veterinarian can decide to inject the pigs with a costly drug that may or may not cure them.

Table 4: Results of the sequential extended oil wildcatter problem with 50 analyzed instances per problem type using the Gurobi parameters in Table 1.

	RDP				RJT				DP			
M	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %
1	0.01	0.00	50/50	+85	0.06	0.04	50/50	+0	0.02	0.01	50/50	+112
2	0.42	0.02	50/50	+271	2.80	0.14	50/50	+0	1.20	0.08	50/50	+358
3	0.77	0.26	50/50	+644	3.95	0.90	50/50	+0	4.01	2.56	50/50	+932
4	50.29	49.77	50/50	+1402	12.05	7.07	50/50	+0	1896	1893	32/50	+2131
5	T	T	0/50	+2934	165.5	119.3	44/50	+0	OoM	OoM	0/50	+4687
6	T	T	0/50	+6009	1249	471.3	32/50	+0	OoM	OoM	0/49	+9968
7	OoM	OoM	0/0	N/A	OoM	OoM	0/0	N/A	OoM	OoM	0/0	N/A
8	OoM	OoM	0/0	N/A	OoM	OoM	0/0	N/A	OoM	OoM	0/0	N/A

Table 5: Nodes of the pig farm problem.

Node	Type	States	Explanation
H_i	Chance node	ill, healthy	Health of the pig in period i
A_i	Chance node	ill, healthy	Test A in period i
B_i	Chance node	ill, healthy	Test B in period i
C_i	Chance node	ill, healthy	Test C in period i
D_i	Decision node	yes, no	Decision to inject the pig with a drug
V_i	Value node	-	Cost of the drug
R	Value node	-	Retail price of a pig

In the following variant, the length of the breeding period varies between 2 and 7 months, and the veterinarian gathers information from three different tests per pig. The influence diagram of the pig farm problem with a four-month breeding period is presented in Figure 2. The nodes of the influence diagram are explained in Table 5.

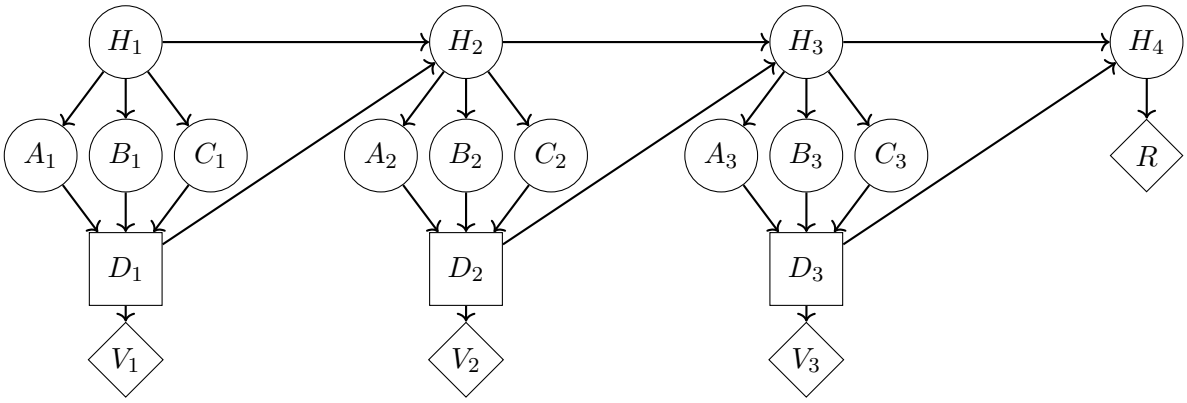


Figure 2: Influence diagram of the pig farm problem with a four-month breeding period.

The pig farm problem represents a case in which the RJT formulation outperforms the RDP formulation computationally. The pig farm influence diagram suits the RJT formulation as it has a periodic structure that results in a very efficient tree decomposition, which forms the basis of the approach. The formulations are tested by solving randomly generated pig farm problems with different breeding period lengths. The problem parameters are generated randomly; the initial probability that a pig is ill is drawn from a uniform distribution Uni$(0, 0.2)$. The sensitivity and specificity of each of the three diagnostic tests are drawn independently from Uni$(0.5, 1)$, which ensures that each test performs better than

Table 6: Results of the pig farm problem with default Gurobi parameters.

	RDP						RJT						DP					
K	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %	# c	# v	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %	# c	# v	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %	# c	# v
2	0.68	0.01	50/50	+7.51	112	288	0.09	0.00	50/50	+0	248	176	10.31	9.20	50/50	+231.4	48	2080
3	1.40	0.55	50/50	+17.60	584	4144	0.02	0.01	50/50	+0	368	262	942.1	937.3	20/50	+499.4	72	7e4
4	1139	1136	48/50	+33.14	4192	7e4	0.03	0.02	50/50	+0	488	348	T	T	0/21	+1092	96	2e6
5	771.0	729.8	4/48	+54.35	3e4	1e6	0.05	0.03	50/50	+0	608	434	OoM	OoM	0/0	N/A	120	7e7
6	OoM	OoM	0/0	N/A	3e5	2e7	0.10	0.02	50/50	+0	728	520	OoM	OoM	0/0	N/A	144	2e9
7	OoM	OoM	0/0	N/A	2e6	3e8	0.05	0.03	50/50	+0	848	606	OoM	OoM	0/0	N/A	168	7e10

Table 7: Results of the pig farm problem with Gurobi parameters from Table 1

	RDP				RJT				DP			
K	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %	$\overline{\tau}_t$ (s)	$\overline{\tau}_s$ (s)	#O/F	RR %
2	0.62	0.01	50/50	+7.28	0.10	0.02	50/50	+0	21.14	20.03	50/50	+228.5
3	1.25	0.47	50/50	+21.9	0.05	0.03	50/50	+0	2229	2223	11/50	+507.5
4	1110	1107	48/50	+34.86	0.07	0.06	50/50	+0	OoM	OoM	0/29	+1056
5	1175	1138	2/2	+57.17	0.08	0.07	50/50	+0	OoM	OoM	0/0	N/A
6	OoM	OoM	0/0	N/A	0.22	0.11	50/50	+0	OoM	OoM	0/0	N/A
7	OoM	OoM	0/0	N/A	0.14	0.12	50/50	+0	OoM	OoM	0/0	N/A

random guessing. The probabilities of the four health state transitions are generated as four independent draws from Uni$(0, 1)$, ordered so that the smallest probability is associated with an untreated ill pig becoming healthy, and the highest probability is assigned to a treated healthy pig remaining healthy. The treatment cost is drawn from Uni$(0, 100)$. The retail price of the pig is determined by drawing two values from Uni$(100, 1100)$ and assigning the higher value as the price of a healthy pig and the lower value as the price of an ill pig.

The results of 50 randomly generated instances for breeding period lengths ranging from 2 to 7 months are reported in Tables 6 (obtained with default Gurobi parameters) and 7 (obtained with Gurobi parameters from Table 1). The computational experiments indicate that the RJT formulation is the most efficient method for finding the optimal decision strategy. The total and solution times and the model size scale linearly with respect to the number of breeding periods. In contrast, for the RDP and DP formulations the total times scale exponentially with respect to the number of breeding periods. This behavior renders the RDP and DP formulations intractable even for problems with moderate breeding period lengths. This example highlights the limitations of the RDP and DP formulations as influence diagrams with a periodic structure, exemplified by the pig farm problem in Figure 2, are more efficiently solved using the RJT formulation.

2.3. Continuous turbine inspection and maintenance problem

Consider the turbine inspection and maintenance problem presented in Section 4.2 of the main manuscript. The influence diagram of the problem is presented in Figure 3. In what follows, assume that the chance nodes described in Table 8 are associated with continuous probability distributions that are specified in Table 9. $\mathcal{N}_{[a,b]}(\mu, [\sigma_1^2, \sigma_2^2])$ denotes the normal distribution with mean μ and variance randomly drawn from Uni(σ_1^2, σ_2^2), truncated on the interval $[a, b]$. $\delta(a)$ is the Dirac delta distribution on a. Table 9 presents the assumed conditional probability distributions. In Table 9, the column *States* specifies the condition that the information state

fulfills for the corresponding probability distribution. For instance, the conditional probability distribution for node TR is $\delta(s_{TE})$ if $s_{IN} = 0$ (i.e., no inspection is conducted). For TR, the variances are randomly generated by selecting two values from $\text{Uni}(0.2, 10)$ and associating the larger variance with the probability distribution conditional on $s_{IN} = 1$. Similarly, the variances for TF are generated by randomly selecting three numbers from $\text{Uni}(0.2, 10)$ and associating the largest variance with the probability distribution conditional on $s_M = 0$ and the smallest variance with $s_M = 2$.

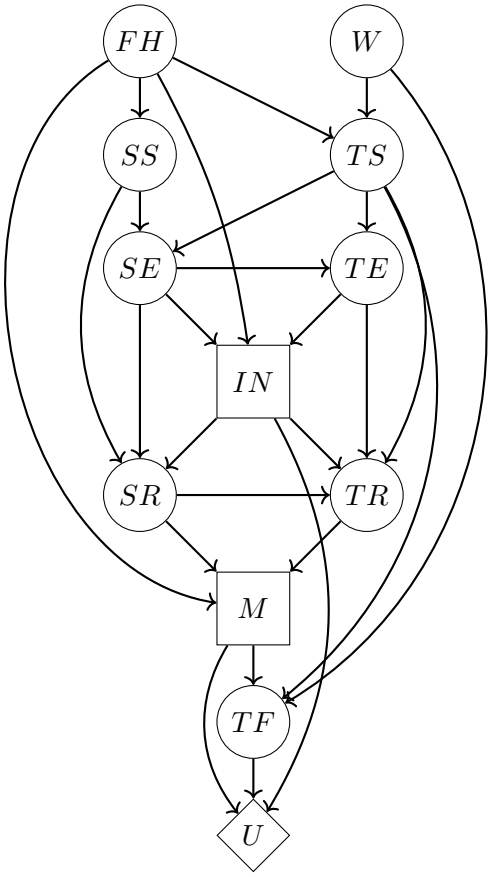


Figure 3: Influence diagram of the turbine inspection and maintenance problem.

Instances of the continuous turbine inspection and maintenance problem are created by randomly selecting the variance of each chance node's probability distribution. Table 9 specifies the range for each variance. Moreover, the costs of the maintenance and inspection alternatives are generated so that the cost of the sensor check is randomly selected from $\text{Uni}(0, 20)$, the cost of turbine inspection from $\text{Uni}(50, 300)$, the cost of level-1 maintenance from $\text{Uni}(100, 2000)$, and the cost of level-2 maintenance from $\text{Uni}(3000, 8000)$. The reward from turbine flow is given by the function $f(s_{TF}) = \frac{1}{100} s_{TF}^3$. The utility associated with consequences is calculated by subtracting maintenance and inspection costs from turbine flow rewards.

Solving the influence diagram representation of the turbine inspection and maintenance problem requires the continuous events to be discretized and represented as a finite set of possible states. To create a discretization of size N for each chance node $c \in C$, $N - 1$ breakpoints $b_n, n \in 1, \ldots, N - 1$ are selected from the domain of the continuous state space.

Table 8: Nodes of the turbine inspection and maintenance problem.

Node	Type	States	Explanation
W	Chance node	[0,100]	Turbine weathering
FH	Chance node	[0,100]	Fired hours
SS	Chance node	[0,100]	True condition of the sensor
TS	Chance node	[0,100]	True condition of the turbine
SE	Chance node	[0,100]	An estimate of the sensor condition
TE	Chance node	[0,100]	An estimate of the turbine condition
IN	Decision node	none, sensor check, turbine check	Decision to inspect the sensor and turbine
SR	Chance node	$[0, 100]$	Result of sensor inspection
TR	Chance node	$[0, 100]$	Result of turbine inspection
M	Decision node	none, level 1, level 2	Maintenance decision
TF	Chance node	[0,100]	Turbine flow after possible maintenance
U	Value node	-	Rewards minus the costs

Table 9: Conditional probability distributions for chance nodes.

Node	Information set	States	PDF
FH	–	-	$\text{Uni}(0, 100)$
W	–	-	$\text{Uni}(0, 100)$
SS	FH	-	$\mathcal{N}_{[0,100]}(100 - s_{FH}, [10, 100])$
TS	FH, W	-	$\mathcal{N}_{[0,100]}(100 - (s_{FH} + s_W)/2, [10, 100])$
SE	SS, TS	-	$\frac{s_{TS}}{100}\mathcal{N}_{[0,100]}(s_{SS}, [10, 100]) + \frac{100-s_{TS}}{100}\text{Uni}(0, 100)$
TE	TS, SE	-	$\mathcal{N}_{[0,100]}(s_{TS}, \frac{100-s_{SE}}{5})$
SR	IN, SS, SE	$s_{IN} = 0$	$\delta(s_{SE})$
		$s_{IN} \in \{1, 2\}$	$\mathcal{N}_{[0,100]}(s_{SS}, 1)$
TR	IN, TS, TE, SR	$s_{IN} = 0$	$\delta(s_{TE})$
		$s_{IN} = 1$	$\frac{s_{SR}}{100}\mathcal{N}_{[0,100]}(s_{TS}, [0.2, 10]) + \frac{100-s_{SR}}{100}\text{Uni}(0, 100)$,
	IN, TS, TE, SR	$s_{IN} = 2$	$\mathcal{N}_{[0,100]}(s_{TS}, [0.2, 10])$
TF	M, TS, W	$s_M = 0$	$\mathcal{N}_{[0,s_{TS}]}(s_{TS} - 0.02s_W, [0.2, 10])$
		$s_M = 1$	$\mathcal{N}_{[s_{TS},100]}(s_{TS} + \frac{(100-s_{TS})}{2} - 0.02s_W, [0.2, 10])$
		$s_M = 2$	$\mathcal{N}_{[s_{TS},100]}(100, [0.2, 10])$

Table 10: Breakpoints used in the turbine inspection and maintenance problem.

N	b_n
2	50
3	50, 75
4	25, 50, 75
5	25, 50, 75 , 87.5
6	25, 50, 62.5, 75, 87.5

The discrete state space is created as $S_c = \{[0, b_1], [b_1, b_2], ..., [b_n, 100]\}$. Table 10 presents the breakpoints used in this example. For simplicity, the same set of breakpoints is used for each chance node. Conditional probabilities are obtained from a set of sample paths S, generated by sampling the conditional probability distributions in Table 9. From the sampled paths, discrete conditional probabilities are obtained as $\mathbb{P}(s_c \mid s_{I(c)}) = |\{\overline{s} \in \overline{S} : \overline{s}_c \in s_c, \overline{s}_{I(c)} \in s_{I(c)}\}|/|\{\overline{s} \in \overline{S} : \overline{s}_{I(c)} \in s_{I(c)}\}|$. For sample paths whose states match $s_{I(v)}$, the utilities $U(s_{I(v)}), v \in V$ are computed as the average utility of those paths.

To highlight the value of the proposed formulation, the performance of the decision strategies generated using discretizations of size $N = 2, ..., 6$ is compared when applied to the turbine inspection and maintenance problem. A total of 400 continuous instances of the problem are randomly generated, and the optimal decision strategies are obtained by solving the influence diagram using discretizations based on the breakpoints in Table 10, following the discretization method described above. In each of the 400 instances, each decision strategy, obtained using a different discretization, is evaluated by sampling the states of the chance nodes 100 000 times, resolving the decision nodes according to the strategy, and recording the resulting utility. We report the average improvement compared to the baseline of using the decision strategy obtained when solving the influence diagram with 2 discrete states. Figure 4 presents the average performance improvement when using a discretization of size N to solve the problem. With the DP and RJT formulations, the largest turbine inspection and maintenance instance that can be solved to optimality can use a discretization of size $N = 4$, which on average improves the performance of the decision strategy by 9.2%. In contrast, the largest solvable instance with the proposed formulation uses $N = 6$ discrete states, yielding an improvement of 11.2%, which is two percentage points more than with the discretization of size $N = 4$. This demonstrates that the RDP formulation can be leveraged to find decision strategies that outperform those obtained with earlier formulations.

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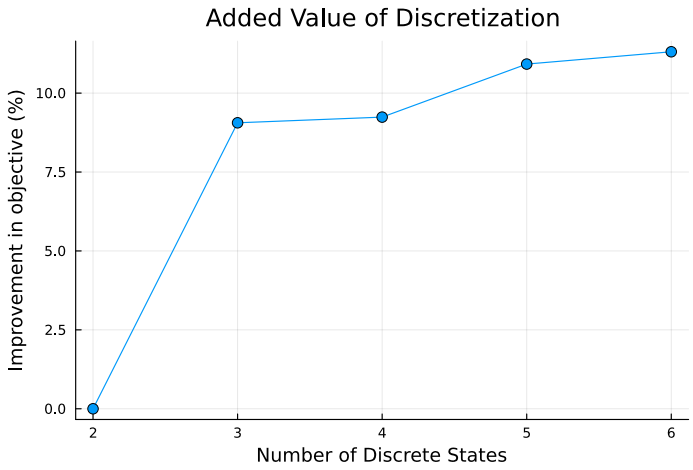


Figure 4: Average relative improvement of optimal decision strategies obtained using a discretization of specific size, compared to the decision strategy obtained when using 2 discrete states.